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(Key and Solutions for AIMCAT2417)

Key

SECTION – I

1. D
2. B
3. C
4. C
5. A
6. C
7. D
8. D
9. C
10. A
11. B
12. B
13. B
14. D
15. C
16. B
17. 1
18. B
19. 4
20. 1342
21. 4213
22. C
23. 4123
24. A

SECTION – II

1. B
2. A
3. B
4. C
5. D
6. 3600
7. 1400
8. 5055
9. D
10. B
11. B
12. C
13. D
14. B
15. A
16. A
17. C
18. 1500
19. 3400
20. C

SECTION – III

1. C
2. 18
3. D
4. C
5. 36
6. C
7. B
8. A
9. B
10. D
11. 3
12. C
13. A
14. 2222
15. B
16. A
17. D
18. C
19. 14000
20. 2
21. B
22. 49

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 448

1. The passage is an objective description of Said's contribution to Post-Colonial studies.
Option A: Choice A is irrelevant/out of scope and cannot be inferred from the bold-faced segments of the text. The statements talk about the western perception of eastern cultures and not the western influence.
Option B: Choice B (internalize colonial assumptions about the inferiority of the conquered peoples) seems contrary to “Said calls into question the underlying assumptions that form the foundation of Orientalist thinking.” “constructing the Orient through the lens of Europeans” contradicts “Scholarship from afar and second-hand representation must take a back seat.” given in the last paragraph.
Option C: The first part of choice C though true [from the term “postcolonial (studies)”] does not emerge from the boldfaced portions of the passage. Also “analyze the western ways of depicting the Eastern world” is the old way of studying the East (as given in paras 2 and 3) which is what is argued against in the last para of the passage. ..... Said calls into question the underlying assumptions that form the foundation of Orientalist thinking. A rejection of Orientalism entails ...... Hence choice C is not correct.
Option D: The passage contextualises Said's work as a background for postcolonial studies. Said's studies both challenge the Western dominance (Scholarship from afar and second-hand representation must take a back seat.) and encourage an Eastern participation in literature about it. ('The Oriental' must be given a voice.).
Choice (D)

2. Option A: The Orient signifies a system of representations framed by political forces that brought the Orient into Western learning, Western consciousness and Western empire. The first

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'Orientalists' were 19th century scholars who translated the writings of 'the Orient' into English, based on the assumption that a truly effective colonial conquest required knowledge of the conquered peoples. Hence choice A is true and is not the answer.

Option B: Rejection of Orientalist thinking ***does not entail a denial*** *of the differences between 'the West' and 'the Orient'*, but rather an evaluation of such differences in a more critical and objective fashion. Hence choice B is false and is the answer.

Option C: What is considered the Orient is a vast region, one that spreads across a myriad of cultures and countries. Said argues that Orientalism can be found in current Western depictions of 'Arab' cultures. ..... A historian and a scholar would turn not to a panoramic view of half of the globe. .... 'The Orient' cannot be studied in a non-Orientalist manner; rather, the scholar is obliged to study more focussed and smaller culturally consistent regions. So choice C is true and is not the answer.

Option D: From "The person who has until now been known as ***'the Oriental' must be given a voice***. Scholarship from afar and second-hand representation must take a back seat to narrative and self-representation on the part of the 'Oriental'", this is true and hence, not the answer.

Choice (B)

3. Refer to the second para.

Option A: Choice A has not been mentioned in the passage.

Option B: Choice B is an exaggeration and does not correctly represent what the sentence in quotes in the question implies.

Option C: The Orient exists for the West, and is **constructed by** and in relation to the West. ***It is a mirror image of what is inferior and alien ('Other') to the West***. Hence choice C is the correct answer.

Option D: While the first part of choice D may be true, the second part is irrelevant in the given context. Also, "defenseless" and "unintelligent" are extreme statements that cannot be inferred from the passage. Hence choice D is not correct.

Choice (C)

4. Option A: In the last para, Said argues that the Orient should be interpreted not as a "panoramic view of half of the globe" but rather as a "focused and complex type of history that allows space for the dynamic variety of human experience". Option A is not the answer.

Option B: From the passage, we see that Said calls into question the underlying assumptions that Orientalist thinking is based on. He mentions that to reject this thinking would entail the rejection of the assumptions about "biological generalizations, cultural constructions, and racial and religious prejudices". Hence, Option B is incorrect.

Option C: In the third para, Said argues that the notions about Orientalism in Western depictions is political propaganda and this is supported by the sermons about "objectivity, the fairness, the impartiality of a real historian, the implication always being that Muslims and Arabs cannot be objective, but that Orientalists ... writing about Muslims are, by definition, by training, by the mere fact of their Westerness.". This is what is perceived by the West and Said disagrees with this in the third paragraph. This option states the exact opposite of the point Said is trying to make. Option C is the answer.

Option D: From the last few lines of the passage, we see that Said talks about how the 'Oriental' should be given a voice and be encouraged to provide a narrative, self-representative view of the Orient, rather than the Orientalist whose "second-hand representation" of the Oriental is gleaned from "scholarship from afar". Hence, Option D is something Said recommends and is not the answer.

Choice (C)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 564

5. In the second para, the author mentions that we are able to distinguish between ourselves, our thoughts and the world.

Option A: The author mentions that we talk about separate the world from us and discuss about it. He also mentions that we can talk about morals and norms. From this we can infer that we can form concepts (for example, about the world) and think about them in abstraction. Hence, this is the correct answer.

Option B: While the author mentions that language helps us discuss our relationship with the world, he does not mention that language is necessary for having social relationships. In the last para also, the author mentions that "No animal possesses either language or a social network like ours". This does not mean that animals do not have any social networks. Hence, this cannot be inferred from the passage.

Option C: The author mentions that because of language, we can "debate, discuss and negotiate among ourselves about the world and our relationship to it". However, the author, when talking about animals mentions that for animals, "their picture of the world is the world". From this we can say that animals also have a picture of the world. The difference between humans and animals is that animals "cannot separate the world as it is from the world as it seems". Hence, this is not the correct answer.

Option D: In the first sentence of the second para, the author mentions that "Because we possess language, we do not simply have experiences, desires and needs, and react to them". From this we can infer that language helps us to do more than just reacting. Also, in the first para, the author mentions that animals, without language, can react to the world. Hence, this option is incorrect. Therefore, the correct answer is option A.

Choice (A)

6. Para 3 says that in the Cartesian World the mind is 'the private possession of the individual'. The author discusses the problem with Cartesian world in the fourth para.
Option A: The problem with Cartesian world is not about the feelings and their expressions. The problem is to do with the meanings of the feelings. Hence, this option is incorrect.
Option B: The author mentions that in a Cartesian world, we can infer that "another human being is in pain because he is acting as I would if I were in pain". This is a possible (but inadequate) answer to the question "How do I know the meaning of pain, as opposed to its sensation?" Hence, this option is incorrect.
Option C: The primary concern with the Carteisan world is that how do we know about the meanings of feelings, if the mind is isolated. The meaning of concepts (like 'self, 'pain') are formed by language (as explained in the next para). Hence, the problem with Cartesian world, as Wittgenstein phrased it, is that "if the only things of which I was certain were the contents of my own mind, then I would not be able to communicate those contents to anyone else". From this, it can be inferred that the Cartesian approach considers the mind to be incapable of communication. Hence, this is the correct answer.
Option D: The author does not discuss about the relation between feelings and thoughts. He only talks about the relation between feelings and the meaning of feelings. Hence, this option is incorrect.
Therefore, the correct answer is option C.
Choice (C)

7. The author presents Wittgenstein's views in paras 5 and 6.
Option A: In the sixth para, Wittgenstein reminds us that "without knowledge of other minds it is impossible to have knowledge of our own". Hence, to know our thoughts and feelings, we need to know about the thoughts and feelings of others. This highlights the importance of social networks and language in learning about ourselves. Hence, this is not a statement that Wittgenstein would disagree with.
Option B: According to Wittgenstein, "our inner our own heads by relating to other humans". From this we can infer that Wittgenstein believes that we can only truly know what goes on inside our own heads by relating to other humans. Inferring other humans' experiences from our own is what Cartesian would have supported. This is also evident from the fourth para, where the author, when discussing the Cartesian world mentions that "We can simply infer that, for instance, another human being is in pain because he is acting as I would if I were in pain". Hence, Wittgenstein would not agree with this statement.
Therefore, the correct answer is option D.
Choice (D)

8. Option A: In the first para, the author mentions that "an animal may be able to react to the world" even without language. Hence, this option is incorrect.
Option B: The author mentions that no animal possesses a language like ours but does not discuss whether animals can be taught the use of language. Hence, this option is incorrect.
Option C: The author mentions that *language is a necessary condition,* ***but not a sufficient condition*** for thinking about the world. Later in the passage, he talks about the importance of social networks and how we understand ourselves by understanding others. From this, it can be inferred that language is not an end in itself for thinking about the world. Hence, this option is incorrect.
Option D: No animal has a language like ours and this is what makes humans self-aware. Hence, this option can be inferred from the passage.
Choice (D)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 541

9. Option A: Wage subsidy is a suggestion made in the passage for creation of local jobs. However, the discussion of greater labour market flexibility is upstream to that, and such flexibility according to the passage has to be coupled with 'the opportunity to stay'. Hence, Option A doesn't give us the factor that makes geographical mobility

feelings are not located entirely inside our heads. They are also the products of our existence as social beings". Also, the last two sentences of the passage highlight the importance of language and social networks. Hence, this is a statement that Wittgenstein would agree with.

Option C: From the last two sentences of the passage, we can infer that since animals lack language and social networks, " it is simply not valid to assume that they have inner experiences as we do". Hence, Wittgenstein would agree with this statement.

Option D: In the last para, the author when providing Wittgenstein's views mentions that "Far from inferring other humans' experiences from our own, we can only truly know what goes on inside

effective.

Option B: From '*Even during times of significant migration, the bulk of local populations stayed put and needed good jobs and solid communities…*', we can understand that the author isn't suggesting migration. Rather, the author is hinting at the past where local populations didn't migrate. The author says that is quite possible and such populations should be allowed to stay and prosper. Hence, Option B is not the answer.

Option C: This option hints at both the factors the author seems to approve of – creation of jobs and building of local communities. Together, these will help in providing people with the 'opportunity to stay'. That will make labour mobility highly effective. Hence, Option C is the answer.

Option D: If that happens, the other side of global mobility is not checked. The passage mentions that geographical mobility should be coupled with the opportunity to stay. So, leaving depressed areas to seek jobs elsewhere isn't the way to go. Rather, there should be creation of jobs and communities for those who want to stay. Hence, Option D is not the answer. Choice (C)

10. Option A: The author brings out the example of a local factory closing to talk about a chain of cascading effects, with people losing jobs and spending less, thus affecting the local economy, eventually leading to other social ills. All this is a result of jobs being outsourced, because of globalisation. Hence, Option A is the answer.

Option B: The author isn't discussing an example of a local factory to delve on social ills and their root causes. Yes, one of the effects of a local factory closing is the loss of jobs and that in turn leads to other social evils. But, the main theme behind the example is to connect communities and globalisation and the effect of the latter on the former. Hence, Option B is not the answer.

Option C: This is the reverse of the situation being discussed in the passage. The author didn't talk about local factories closing down to bring out something positive. It is a negative example, and the reverse may or may not work out to be true.

governments should focus on that. How that can be made to happen is a downstream idea and hence, is not the central theme of the passage. Option C is not the answer.

Option D: Once again, the option deviates from the main idea of investing in local communities to discuss economies and how they will shape up. Once local communities are strong, what the benefits are is an idea not discussed in detail in the passage. Also, the author didn't really compare the pros and cons of globalisation and building of local communities, against each other. Hence, Option D is not the answer. Choice (B)

12. Option A: It has been mentioned in the first para that pride in one's culture and history is an essential element of a prosperous society, which contributes to the health of a local community. This can be understood from '*pride in local cultures and history – these are the essential elements of prosperous societies.*' Hence, Option A is not the answer.

Option B: While our current version of economics doesn't focus on the health of local communities, it is not possible to infer what it focuses on. It is also not possible to compare focus on markets vis-à-vis prosperity of individuals (as opposed to societies). Also, we cannot infer from the first para that the market is the counter to local community,

Hence, Option C is not the answer.
Option D: Communities and local economies are interdependent and together need stable jobs to prosper. That's what the author is trying to elucidate. It isn't to make a point that it is communities that build local economies (which deteriorate along as the spending power of people goes down due to loss of jobs). Hence, Option D is not the answer. Choice (A)

11. Option A: The central argument of the passage is that local communities must be encouraged to prosper. It is not about how 'instability' of local environments affects 'economies'. Economies is not the primary point of the passage. It is about building prosperous local communities (not just because it is important for economies, since economy is a much smaller idea compared to the importance given to communities). Option A is not the answer.
Option B: This is the main idea of the passage as understood from the following lines: *'Two books, one by Raghuram Rajan and another by Oren Cass, revisit our economistic worldview and argue that* ***we should instead put the health of our local communities front and centre.*** ' And: *'How we balance these forces with the needs of communities will shape not only our economic fortunes, but also our social and political environment. As Cass and Rajan show,* ***it is a problem that economists should no longer ignore***.' Option B is the answer.
Option C: This option talks about relationship between jobs and communities. The main argument is to help local communities prosper and given that Rajan talks about three pillars, and not two. Hence, Option B is the answer.
Option C: This can be understood from the sentence: '*Neither global markets nor the nation-state can adequately supply them.*' In other words, the markets don't focus on the prosperity of local communities and have an entirely different function (not mentioned in the passage). Option C is not the answer.
Option D: The last line of the first para clearly indicates that global markets and the nation-state ***sometimes undermine 'them'***, them being the essential elements of prosperous societies, which could be equated to all the factors (stable jobs, culture, history, etc.) that contribute to the health of local communities. Hence, Option D is not the answer. Choice (B)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 537

13. The main purpose of the author in the passage is to explain the effects of bilingualism on the brain.
Option A: The author does not want to just address the benefits of bilingualism. The author wants to evaluate if bilingualism offers advantages as has been perceived. So, choice A is incomplete or narrow in scope.
Option B: Throughout the passage, the author discusses the extent of the benefits of bilingualism on the brain – if it actually offers any advantages, if bilinguals actually have an upper hand over

monolinguals etc. The author's conclusion is mentioned at the end of the passage. The bilingual advantage may not appear in the exact guise researchers think of it today. But the advantages may be different from the way researchers describe it. Hence, this is the answer.

would be: Bilinguals have an advantage in terms of a quicker reaction time when the right-pointing arrow appears on the left and the left-pointing arrow appears on the right – these are the incongruent trials. Choice A is not the answer.
Option B: Since it doesn't matter on which side of

Option C: The primary concern of the author is not to prove that bilingualism is more advantageous than monolingualism. Throughout the passage, the author cites the results of experiments/studies done which tell us that the bilingual advantage is sometimes overstated. Refer to the first two sentences of the third para: When de Bruin looked at the data, though, in three of the four tasks testing inhibitory control, including the Simon task, the advantage wasn't there. Monolinguals and bilinguals had performed identically. Refer to the first two sentences of the fourth para: De Bruin isn't refuting the notion that there are advantages to being bilingual. But the advantage is neither global nor pervasive. Hence, this is not the answer.
Option D: The advantages that a bilingual child may or may not enjoy has been discussed at the end of the passage. De Bruin adds that the actual advantages of bilingualism may be different from these. However, choice D is not the primary concern of the author. Choice (B)

14. In 2012, Angela de Bruin, a bilingual, pursued the link between bilingualism and cognition. When de Bruin looked at the data, though, in three of the four tasks testing inhibitory control, including the Simon task, the advantage wasn't there. Monolinguals and bilinguals had performed identically.
Option A: De Bruin does not intend to refer to the reliability of the experiment. The literal implication of 'reliability' does not emerge in the case of the experiment. In her statement, she refers to “existing literature” and not the data from her experiment. Hence, choice A is incorrect.
Option B: Choice B is besides the point and is not the reason for De Bruin's statement.
Option C: Choice C speaks about the result of the experiment, but the result of the experiment was not inconclusive as mentioned in this option. It was conclusive but it did not yield the expected results. Hence, this does not answer the question.
Option D: “The existing literature is not a full, reliable picture” means that the existing literature is not complete. Hence, the data was insufficient to conclude that bilingualism is advantageous. Hence, de Bruin decided to test it further. So, she combed through conference abstracts from 169 conferences that dealt with bilingualism and executive control. Choice D is the answer.
Choice (D)

15. In the Simon task (discussed in para 2), if you see a right-pointing arrow, you press the right key. It doesn't matter on which side of the screen the arrow appears but only the direction in which it points.
Option A: Choice A is the opposite of the findings mentioned in para 2. The correct observations the screen the arrow appears, choice B is similar to choice A and the same explanation given for choice A holds true for choice B as well. Choice B is not the answer.
Option C: Choice C is correct. The incongruent trials here would be: upward-pointing arrow appears at the bottom and a downward-pointing arrow appears on the top. Only bilinguals are perceived to have an advantage in the case of this incongruent trial.
Option D: Choice D states the converse of what has been correctly mentioned in choice C. Hence choice D cannot be the answer. Choice (C)

16. The study mentioned in para 4 aimed to test where the limits of bilingual advantage lie, and what the real advantage may actually look like. The findings of the study have been mentioned in the next para i.e. para 5. The fourth task which needs to be considered is the simple task-switching paradigm (you see circles and squares that are either red or blue, and must pay attention to either one color or one shape, depending on the part of the trial).
Option A: On the last task, they thought they'd finally detected an advantage: on the switch trials – the trials immediately after a change from shape to color or color to shape – the bilinguals, both active and passive, **seemed** to be quicker. **But** when the researchers dug deeper, they found that it wasn't so much a case of **switching faster** as it was being **slower at the non-switch trials, where shape followed shape and color followed color**. While choice A can explain the result of the switch trials, it cannot explain the result of the non-switch trials.
Option B: But when the researchers dug deeper, they found that it wasn't so much a case of switching faster as it was being slower at the non-switch trials, where shape followed shape and color followed color. i.e. In switch trials, bilinguals were not faster than monolinguals. But in non-switch trials, bilinguals were slower than monolinguals. On the whole we can infer that bilinguals were not as fast as monolinguals in this test. If choice B is true, then it will corroborate the results of the test.
Option C: The bilinguals, both active and passive, **seemed** to be quicker. **But** when the researchers dug deeper, they found that it wasn't so much a case of **switching faster** as it was being **slower at the non-switch trials, where shape followed shape and color followed color**. Now, if choice C were true, then bilinguals should have shown a distinct advantage over monolinguals in the study.
Option D: The Simon task involves pictures and researchers found no difference between monolinguals and bilinguals. Even in the third task i.e. the Tower of London (you solve a problem by

moving discs around on a series of sticks to match a picture of what the final tower looks like), the researchers found no difference between monolinguals and bilinguals. Choice D does not support the findings of the study.

Choice (B)

Solution for question 17:

17. Sentence 1: Sentence 1 has the adverb 'indeed'. It is negative in tone.

Sentence 2: Sentence 2 begins with the conjunctive adverb 'however'. It mentions a conclusion of some research.

Sentence 3: Sentence 3 has a reference to 'that assertion'.

Sentence 4: Sentence 4 mentions a supposition. It sounds like an ideal introductory sentence.

Sentence 5: Sentence 5 has the adverb 'indeed'. It talks about how people leaving university have more inquiring minds than those who work or engage themselves in vocational training.

Between sentences 4 and 5, sentence 4 is a more general sentence that can begin the para. Sentence 4 talks about 'going to university'. Sentence 5 compares the benefits of 'going to university' vis-a-vis 'vocational training for work'. So there has to be a sentence introducing 'training for work' before sentence 5.

Hence sentence 4 begins the para. It mentions the supposition: university education is a mind-broadening experience. Sentence 4 is followed by sentence 3. "That assertion" in sentence 3 points to the supposition given in sentence 4. We understand from sentences 4 and 3 that "going to university" is a different experience than "training for work". "supposed to be a mind-broadening experience" in sentence 4 contrasts "might not be so stimulating" in sentence 3.

Sentence 3 is followed by sentence 5. "do indeed seem" in sentence 5 reiterates the point made in sentence 3. "seem to leave with broader and more inquiring minds" in sentence 5 points to "mind-broadening experience" in sentence 4 and contrasts "might not be so stimulating" in sentence 3.

Sentence 5 is followed by sentence 2 which concludes the para. "it was not the case that university broadened minds; rather, work seemed to narrow them" is an outcome that contradicts the supposition "supposed to be a mind-broadening experience" in the opening sentence 4. So, 4352.

Sentence 1 is the odd sentence out. It needs a precedent and more substantiation. It runs tangent to the thought flow. Ans: (1)

Solution for question 18:

18. Option A: The paragraph states that despite their philanthropic nature, companies managed to be profitable. It does not imply that they are profitable because of a belief system of social responsibility. It also does not mention philanthropists being concerned about the profitability of their companies. Hence, this is not the answer.

Option B: From the penultimate sentence of the paragraph, twenty-five companies in Estonia have been actively engaged in CSR regulation and public policy development in the 21st century and are **yet**, very profitable. Hence, choice (B) can be inferred.

Option C: The latter half of choice (C) defines philanthropy but the first part is a wrong cause-effect sequence and cannot be inferred.

Option D: It is mentioned in the paragraph that companies philanthropy, tendency to punish corrupt directors and employees is because of their belief systems and proactive actions. This need not simply mean belief in social responsibility. Besides, this statement assumes that companies are philanthropic first and then have the tendency to punish corrupt directors and employees. But the paragraph says the philanthropy as well as tendency to punish both arise from their belief systems. Hence, this is not the answer. Choice (B)

Solution for question 19:

19. Sentence 1 talks about the origin/history of martinis connecting it to the gold rush in California.

Sentence 2 talks about how martinis didn't come from 'gentlemen's clubs' but from a rough and tumble side. So, this sentence sits with the first sentence because the first talks about the humble origins of martini – what's explained in 1 is more akin to the rough and tumble side rather than the gentleman's clubs.

Sentence 3 talks about the genesis of a drink and hence can only follow an in-progress recipe.

Sentence 4 talks about 'another account' and this time the story is different from the California recipe.

Sentence 5 talks about recipe spreading through California, which connects 5 to 1.

1 and 5 are connected, and 5 directly leads to 3, (lost the 'ez' and the sweet liqueur and a race car of a drink was born). 2 is upstream of 1 while 4 is downstream, a different account and possibly part of another para. So, 2 is more relevant to the para. The order is 2153. The odd one out is sentence 4.

Ans: (4)

Solutions for questions 20 and 21:

20. Sentence 1 describes the difference between the roles of women in agricultural extension in a bunch of countries.

Sentence 2 is a direct reference to some sentence that discusses Indonesia.

Sentence 3 offers a reason for 'the difference' and the reason is based in quite a few things including religion.

Sentence 4 talks about Thailand and Philippines society and how they are matrilineal, drawing a contrast to Indonesia where women's role in agricultural extension is limited because of the restrictions of the religion.

We can understand that the difference discussed in 3 is that in 1 and not in 4, because 4 attributes the difference purely to religion, and hence a more specific idea compared to Sentence 3. 3 on the other hand discusses several ideas, therefore, more generic than 4. So, 13 is a block. Similarly, 2 needs 4 since 2 refers to overcoming the problem in Indonesia. So, 42 is a block. Hence, 1342. Ans: (1342)

21. Sentence 1 talks about Kobe and Jordan, but only half names, with other sentences using full names. So, 1 has to be preceded by other sentences. We also understand that Kobe is referred to as the Black Mamba.
Sentence 2 uses 'their' which means the preceding sentence has to reference more than one person.
Sentence 3 has the connector 'also' which means MJ's message in this sentence has to be preceded by something else MJ has said. So, 3 has to follow 1 which is about MJ's first message, that Kobe is a worthy adversary.
Sentence 4 is an independent sentence that introduces the two guys.
2 has to follow 4 because it explains the theory in 4. We see a spitting image of MJ in Kobe because their offensive games are eerily similar. So, 4213 is the order. Ans: (4213)

Solution for question 22:

22. Option A: This need not be checked because the para itself says there was a burst of evolution and new biodiversity. So, unless we are saying the para is not true (obviously, can't say that) we don't really need to verify this information. Option A is not the answer.
Option B: Biodiversity increase means there were new species. Period. We don't really need to know whether the new species were drastically different, or more or less similar to the other. Even if we did, even if there were differences, it won't tell us whether the burst was organic and long-time coming or whether it got a spark from somewhere.
Option C: This will answer the question that there was a spark. If there was a change in atmospheric conditions coinciding with the burst of biodiversity, then we know that there could be a spark. So, the answer to this question, doesn't definitively prove that there is a spark, but helps us further evaluate the claim that there was indeed something that sparked the sudden growth of biodiversity and appearance of new species. (We cannot be sure because it could be correlation, not causation, that it was just a coincidence, but without this question, in marine habitats or not doesn't prove that something sparked the sudden growth in biodiversity. Option D is not the answer.
Choice (C)

Solution for question 23:

23. On a careful reading of the sentences, it can be observed that sentence 4 is a general sentence that begins the paragraph. It mentions an annoyance that a researcher experiences in the area of scientific publishing: slowness. Sentences 4 and 1 form a mandatory pair. "slowness in **scientific publishing** will come near the top of the list of gripes" in sentence 4 links with "It takes nearly six months ..... wend its way from submission to **publication**" in sentence 1 and a paper is often rejected by one or more others" in So, 41. Sentence 1 is followed by sentence 2. "rejected by one or more others" in sentence 1 is explained by "The reason need not be a fatal flaw in the research; sometimes the work is simply not splashy enough" in sentence 2. Sentence 3 follows sentence 2 and concludes the paragraph. Hence, 4123. Ans: (4123)

Solution for question 24:

24. Option A: An answer to this question will help us evaluate the argument because if it is true that there is evidence for greater hunting activity, it strengthens the hypothesis that there was a shift to animal protein. Similarly, scavenging leads us to a similar result. The only minor assumption we have to make is that hunting and scavenging was for food, a very acceptable assumption to make. Hence, Option A is the answer.
Option B: Irrespective of what the answer to this is, we cannot correlate an argument about dietary change to tools (because we cannot make assumptions about how those stone tools were used or why they were used). Hence, Option B is not the answer.
Option C: The information presented in the paragraph talks about animal protein and other high energy food resources. The two need not be the same. Hence, Option C is not the answer.
Option D: The argument doesn't establish a relationship between calories and digestion and, hence, knowing whether the metabolic rates went up or not will not give us much to evaluate whether the dietary patterns shifted to animal protein (we cannot assume that more the protein or high-energy source consumed, more the metabolic rate). Option D is not the answer.
Choice (A)

we won't even know if there was a spark). Option C is the answer.
Option D: Even if this were true, it won't prove that the burst of evolution happened because of a spark, because we are still not sure, according to the para, that rising sea-levels flooded low-lying areas. So, when we don't know if there were marine habitats, knowing whether life-forms thrive

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	9, 10, 13, 14, 15, 16, 17, 18, 21
Difficult	2, 4, 5, 6, 7, 8, 11, 12, 19, 23, 24
Very Difficult	1, 3, 20, 22

SECTION – II

Solutions for questions 1 to 5:

From (iii), C met E but did not meet B. Further, C met exactly 3 persons.
From (iii) and (iv), D must have met either 4 or 5 persons (since D met more number of persons than C).
From (v), E met both B and F. Since G did not meet any person that F met, G did not meet E.
Since F met E, from (i), A must also have met E.
4 persons have definitely met E. E must have met a total of 4 or 5 persons. Hence, E and D must have met 4 and 5 persons in any order.
If E met only 4 persons, then E and D must not have met each other. Hence, D must have met five persons, i.e., all the persons apart from E. However, in this case, D must have met both F and G which violates condition (v). Hence, this is not possible.
Hence, E must have met 5 persons (including D), while D must have met 4 persons.
Since the number of persons that the six persons met are 1, 2, 2, 3, 3, 4 and 5, and E and D have met 5 and 4 persons respectively, B must have met 1 or 2 or 3 persons.
However, A did not meet two persons that B met. A and B together already have met E. Hence, apart from E, B must have met at least two other persons, whom A must not have met. Hence, B must have met a minimum of 3 persons. Since the maximum number of persons that B can meet is 3, B must have met exactly 3 persons. Apart from E, the other two persons that B met cannot be met by A.
D must have met 4 persons and has met E. Between F and G, D can meet at most one person (from (v)). Between A and B, D can only meet one person. Hence, D must have met C.
Since C met 3 persons, and has met D and E, C must have met one more person. If C met F, then from (i), A must also have met C. Since this is not possible, C must not have met F. Between A and G, C must have met one person. Since A and B did not meet, B could not have met F (from (i)). Hence, B must have met both D and G. Since B met D, A cannot meet D. Since A and D did not meet, D and F cannot meet. Hence, D must have met G. Since G met 2 persons, B and D, G cannot meet anyone else.
Since G and C did not meet, C must have met A. Since A met 2 persons, A and F must not have met.
The following table presents the persons that each person met:

	A	B	C	D	E	F	G	No. of Persons met
A	X	X	✓	X	✓	X	X	2
B	X	X	X	✓	✓	X	✓	3
C	✓	X	X	✓	✓	X	X	3
D	X	✓	✓	X	✓	X	✓	4
E	✓	✓	✓	✓	X	✓	X	5
F	X	X	X	X	✓	X	X	1
G	X	✓	X	✓	X	X	X	2

1. G met exactly 2 persons. Choice (B)

2. E met the highest number of persons. Choice (A)

3. A met C. Choice (B)

4. B met G but C did not meet G. Choice (C)

5. Both E and D together met only two persons, B and C. Choice (D)

Solutions for questions 6 to 10:

From the given condition, we can see that the company ranked 4th will have twice the revenue of the company ranked 8th, thrice the revenue of the company ranked 12th, and so on until seven times the revenue of the company ranked 28th.

Similarly, the revenue of the company ranked 5th will have twice the revenue of the company ranked 10th, thrice the revenue of the company ranked 15th, and so on until six times the revenue of the company ranked 30th.

The revenue of the company ranked 6th will have twice the revenue of the company ranked 12th, thrice the revenue of the company ranked 18th, and so on until five times the revenue of the company ranked 30th.

This process can be continued for all possible values of n.

Let the revenue of the company ranked 4th to be 4200 (which is the smallest number divisible by all the numbers from 1 to 7).

The companies ranked 8th, 12th, 16th, 20th, 24th and 28th will have revenues of 2100, 1400, 1050, 840, 700 and 600 respectively.

Since the revenue of the company ranked 20th is 840, the revenues of the company ranked 5th must be 4 × 840 = 3360.

From this, the revenues of the companies ranked 10th, 15th, 25th and 30th are 1680, 1120, 672 and 560 respectively.

Since the revenues of the company ranked 30th is 560, the revenues of the company ranked 6th must be 5 × 560 = 2800.

From this, the revenues of the companies ranked 18th is $\frac{2800}{3}$.

The revenues of the company ranked 9th must be $2 \times \frac{2800}{3} = \frac{5600}{3}$.

The revenues of the company ranked 27th must be $\frac{5600}{9}$.

Since the revenues of the company ranked 28th is 600, the revenues of the company ranked 7th must be 4 × 600 = 2400.

The revenues of the company ranked 14th and 21st must be 1200 and 800, respectively.

The following table provides the relative revenues of all the companies that we calculated:

Rank	Revenue	Rank	Revenue	Rank	Revenue
1		11		21	800
2		12	1400	22	
3		13		23	
4	4200	14	1200	24	700
5	3360	15	1120	25	672
6	2800	16	1050	26	
7	2400	17		27	622.22
8	2100	18	933.33	28	600
9	1866.67	19		29	
10	1680	20	840	30	560

Since we have the relative revenues of all the companies ranked 6 to 10, we can find the sum of these revenues. The sum of relative revenues of companies ranked 6 to 10 (from the relative revenues) = $2800 + 2400 + 2100 + \frac{5600}{3} + 1680 = \frac{32540}{3}$.

The average revenue given (from the table) for these 5 companies is 9762. The total revenue of these five companies is 9762 × 5 = 48810.

The ratio of the value that we calculated and the value that is given = $\frac{48810}{\frac{32540}{3}}$ = 4.5

Hence, the revenues of all the companies that we have considered must be multiplied by 4.5.

6. The revenue of the company ranked 21 = 800 × 4.5 = 3600. Ans: (3600)

7. The revenue of the company ranked 18 = 2800/3 × 4.5 = 4200.
The revenue of the company ranked 27 = 5600/9× 4.5 = 2800.
The required difference = 1400. Ans: (1400)

8. The average of the revenues of the companies ranked 14, 15 and 16 = $\frac{1200+1120+1050}{3} \times 4.5 = 5055$ Ans: (5055)

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9. The total revenue of the companies with ranks 11 to 15 is 5996.8 × 5 = 29984.
The total revenue of the companies ranked 11 and 13 = 29984 – (1400 + 1200 + 1120) × 4.5 = 13244.
The revenue of the company ranked 11 must be an even number (since the revenue of the company ranked 22 is also an integer). Also, the revenue of the company ranked 22 cannot be more than that of the company ranked 21st, i.e., 800 × 4.5 = 3600. Hence, the maximum possible revenue of the company ranked 11th is 3600 × 2 – 2 = 7198.

180/6 = 30 ml for manufacturing one widget in Turbo mode.
Machine C manufactured first 180 widgets in normal mode and the next 820 widgets in turbo mode.
Total fuel consumed by machine C for manufacturing the first 1000 widgets = 180 × 15 + 820 × 30 = 27300.
Machine E manufactured first 60 widgets in Normal mode and the next 540 widgets in Turbo mode, the next 360 in Normal mode and the next 40 in Turbo mode.
Total fuel consumed by machine E for

Similarly, the revenue of the company ranked 26th cannot be more than the revenue of the company ranked 25th, i.e., 672 × 4.5 = 3024. Hence, the maximum possible revenue of the company ranked 13th is 3024 × 2 – 2 = 6046.
Since the sum of these two values is 13244, **the revenues of the companies ranked 11th and 13th** must be 7198 and 6046, respectively.
The revenues of the companies ranked 22nd and 26th must be 3599 and 3023 respectively.
The revenue of the company ranked 11th = 7198.
Hence, option D is the answer. Choice (D)

10. The total revenue of the companies ranked 20 to 25 is 3359 × 5 = 16795.
From the above solution, we know the revenue of the companies ranked 22nd and 26th.
The revenue of the company ranked 23rd = 16795 – 3600 – 3599 – 3150 – 3024 = 3422.
The total revenue of the companies ranked 26 to 30 = 2734 × 5 = 13670.
The revenue of the company ranked 29th = 13670 – 3023 – 2800 – 2700 – 2520 = 2627.
Required difference = 3422 – 2627 = 795
Choice (B)

Solutions for questions 11 to 15:

11. From the given information, we can see that in normal mode, for every 30 minutes, 60 widgets will be produced and in turbo mode, every 30 minutes, 180 widgets will be produced.
From the graph, we can count the number of widgets produced every 30 minutes, based on whether the machine is operated in normal mode or turbo mode.
The number of widgets manufactured by machine A = 120 + 720 + 180 + 540 + 180 + 360 + 60 = 2160
The number of widgets manufactured by machine C = 180 + 1080 + 180 + 720 + 120 = 2280.
Required difference = 2280 – 2160 = 120.
Choice (B)

12. The lowest fuel will be consumed by the machine that ran for the least time in Turbo mode. By observation, we can see that machine E ran the least time in Turbo mode. Hence, the answer is machine E. Choice (C)

13. Any machine consumes 30/2 = 15 ml for manufacturing one widget in Normal mode and manufacturing first 1000 widgets = 60 × 15 + 540 × 30 + 360 × 15 + 40 × 30 = 23700.
Required difference = 3600 Choice (D)

14. In Normal mode, any machine can produce 2n widgets every n minutes.
In Turbo mode, any machine can produce 6n widget every n minutes.
For a machine to produce at least 4n widget in n minutes, it has to run for in Turbo mode for at least as much duration as in Normal mode until the end of n minutes. This condition has to be satisfied by all the machines.
From the graph, we can see that the times at which the condition is satisfied for Machine A is from 10:00 to 12:00, 13:00 to 15:00, 16:00 to 17:00.
For Machine B, the times are from 10:30 to 13:30, 15:00 to 17:00.
For Machine C, the time is from 11:00 to 17:00.
For Machine D, the times are from 8:00 to 14:00, 15:00 to 17:00.
For Machine E, the time is from 9:00 to 11:00.
Hence, only at 11:00 AM is the given condition satisfied for all the machines.
The only possible value of n = 180. Choice (B)

15. Any machine consumes 30/2 = 15 ml for manufacturing one widget in Normal mode and 180/6 = 30 ml for manufacturing one widget in Turbo mode.
For a machine to consume 25 ml of fuel for each widget that it manufactures, the number of widgets that it has to manufacture in Normal mode and Turbo mode should be in the ratio ½ or lower.
Since the times taken to manufacture each widget in Normal mode and Turbo mode are 30 seconds and 10 seconds, the duration for which the machines should be in Normal and Turbo modes must be in the ratio 3/2 or lower.
To find the minimum possible value of n, we need to find the earliest time at which the ratio will be 3/2 or lower.
Machine B was in Normal mode from 8:00 to 9:15, for 75 minutes.
It then switches to Turbo mode. For the ratio of normal and turbo mode to be 3/2 or lower, the machine should be in Turbo mode for at least 50 minutes.
Hence, in the first 75 minutes, Machine B will manufacture 150 widgets. Fuel consumed for this is 150 × 15 = 2250 ml.

In the next 50 minutes, in turbo mode, Machine B will manufacture 300 units. Fuel consumed for this is 300 × 30 = 9000 ml.

Hence, the minimum possible value of n is 11250.
Choice (A)

∴ After manufacturing 450 widgets, Machine B would have consumed 11250 ml (which is 450 × 25).

Solutions for questions 16 to 20:

Let P1 to P8 represent the positions around the circular table in the clockwise direction, with P1 at the top.

From (iii), A played in the sixth round (since A was still sitting opposite H at the end of the fifth round). From (v), the two persons who played in the seventh round were sitting adjacent each other. Since the two chairs adjacent to A are both empty (from (iii)), A could not have played in the seventh round. Hence, A must have been knocked out in the sixth round.

H must have played in the seventh round.

Also, from (v), the last two rounds were won by different persons. This is possible only if one of the two persons who played the last round won the sixth round and the other won the seventh round.

From (ii), both C and E must have played in the fourth round. C and E could not have been knocked out in the first three rounds.

Also, from (vii), F must not have been knocked out in the first five rounds. A and H also would not have been knocked out in the first six rounds.

Hence, the persons who could have been knocked out in the first three rounds must be B, D and G, in any order.

From (i), D must have won at least one round. Hence, D could not have been knocked out in the first round.

From (iv), B could not have been knocked out in the third round. D also could not have been knocked out in the third round because the person who was knocked out in the third round must not have won any rounds. Hence, the only person who could have been knocked out in the third round is G. Hence, D must have been knocked out in the second round and B must have been knocked out in the first round.

Also, D must have won the first round and must have had 250 (initial amount) + 500 (amount with B) = 750 at the end of the first round.

In the fourth round, either C or E must have been knocked out. However, from (ii), C must win at least one round and C cannot win first round (since D won it) and cannot win second round (since amount with C will become more than ₹1200) and cannot win third round (since G, who was knocked out in the third round, has 800, which will take C's total amount to more than 1200).

Hence, C could not have been knocked out in the fourth round. E must have been knocked out in the fourth round.

In the fifth round, both C and F played (along with A and H). One of C and F must have been knocked out and the other must have played in the seventh round.
If C played in the seventh round, he must have won either the sixth round or the seventh round. Since he did not have more than ₹1200, he could not have won the seventh round (as he would have won the total amount with every person). If C won the sixth round, C will have A's amount and his amount, which adds up to more than ₹1200. Hence, C could not have played in the seventh round.

C must have been knocked out in the fifth round. Since C must win at least one round, C must have won the fourth round. Also, E could not have won the second or third rounds since C's amount will become more than ₹1200 when E's amount is transferred to C.

From (vii), F must have won the sixth round and H must have won the last round. The second and third rounds can only be won by A or F or H. From (vii), H won at most one round. Hence, H cannot win the second or third rounds.

Let A be sitting at P1. From (iii), H must be sitting at P5. From (v), F must be sitting adjacent to H. Also, from (vi), B must be sitting to the left of F. Hence, F must be in P6 and B must be in P7.

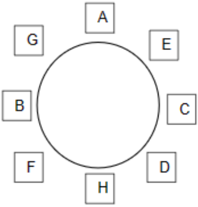
From (ii), C and E must be adjacent each other. They cannot be in P8. Also, from (i), D cannot be in P8. Hence, G must be in P8.

From (vi), the person who won the second round cannot be A because A was sitting opposite H, who won one round. Hence, F must have won the second round. Also, the person sitting opposite F can only be E, since C and D have won one round each. Hence, E must be in P2. From (ii), C must be in P3 and D must be in P4.

Also, from (viii), A must have won the third round. Also, since F won the sixth round, A must have won the fifth round (H cannot win this round as well since he won only one round).

The following table provides the persons knocked out in each round, the persons who won each round and the persons who played each round. It also provides the amounts with each person at the end of that round. The figure provides their positions around the table:

Round	**Knocked Out**	**Winner**	**Amounts**							
			A	B	C	D	E	F	G	H
1	B	D	700	0	600	750	150	400	800	550
2	D	F	700	0	600	0	150	1150	800	550
3	G	A	1500	0	600	0	150	1150	0	550
4	E	C	1500	0	750	0	0	1150	0	550
5	C	A	2250	0	0	0	0	1150	0	550
6	A	F	0	0	0	0	0	3400	0	550
7	F	H	0	0	0	0	0	0	0	3950



16. F won the second round of the tournament.
Choice (A)

17. H had the lowest amount at the beginning of the fifth round.
Choice (C)

18. A had ₹1500 at the end of the third round.
Ans: (1500)

19. F had ₹3400 before the beginning of the seventh round.
Ans: (3400)

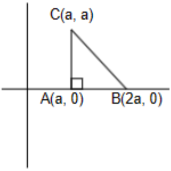
20. C won the fourth round. B was sitting opposite C.
Choice (C)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	11, 12
Medium	1, 2, 3, 4, 5, 13, 14, 15
Difficult	6, 7, 8
Very Difficult	9, 10, 16, 17, 18, 19, 20

SECTION – III

Solutions for questions 1 to 3:

1.



We can see that AB = AC = a and ∠CAB = 90°

∴ ABC is a right-angled isosceles triangle

∴ Its area = $\frac{a^2}{2} = 16$

$\Rightarrow a = \sqrt{32} = 4\sqrt{2}$ (as $a > 0$)

Choice (C)

2. The minimum value of $\frac{(x+a)(x+b)}{(x+c)}$ is obtained when $x = \sqrt{(a-c)(b-c)} - c$

In $\frac{(x+4)(x+10)}{x+2}$, a = 4, b = 10, c = 2

$X = \sqrt{(4-2)(10-2)} - 2$

$= \sqrt{(2)(8)} - 2$

$= 4 - 2 = 2$

$\therefore$ The minimum value is $\frac{6 \times 12}{4} = 18$

Alternative Solution:

$$\frac{(x+4)(10+x)}{x+2} = \frac{x^2+14x+40}{x+2}$$

$$= \frac{(x^2+4x+4)}{x+2} + \frac{(10x+36)}{x+2}$$

$$= x+2+\frac{10(x+2)}{x+2}+\frac{16}{x+2}$$

$$= 10+\left[(x+2)+\frac{16}{(x+2)}\right]$$

Now, the product of the above two terms in the brackets is a constant and equal to 16. Hence, the sum is the minimum when each term is equal to the square root of the constant product, i.e., 16.

Hence, $(x+2) = \frac{16}{(x+2)} = \sqrt{16} = 4$

Hence, required minimum of $\frac{(x+4)\,(10+x)}{x+2}$

= 10 + [4 + 4] = 18

Ans: (18)

3. Since the four expressions (say A_1, A_2, A_3 and A_4) are in A.P, the sum of the first and the fourth expressions equals the sum of the second and the third expressions

$\Rightarrow (a_1 - 2a_2) + (4a_4 - a_1) = (2a_2 - 3a_3) + (3a_3 - 4a_4)$

$\Rightarrow a_2 = 2a_4$ ----- (1)

Also, $A_2 - A_1 = A_3 - A_2$

$\Rightarrow (2a_2 - 3a_3) - (a_1 - 2a_2) = (3a_3 - 4a_4) - (2a_2 - 3a_3)$

$\Rightarrow 8a_2 - 9a_3 = a_1$ ----- (2)

Solutions for questions 4 and 5:

Let the 6-digit number be *abcdef*. Given that *a*, *c*, *e* are odd and *ab*, *cd*, *ef* are multiples of 8.

$\therefore$ *ab*, *cd*, *ef* can be 16, 32, 56, 72 or 96.

To get a 6-digit number of the required kind, we have to arrange 3 of these numbers, which can be done in $^5P_3$ ways, 5(4) (3) or 60 ways.

4. As there is only one 6, there is only one group among 16, 56, 96.

$\therefore$ Both 32 and 72 have to be there. Thus, the digits which have to be there are 2, 3, 7 and 6. Among the options, only 7 occurs. Choice (C)

5. *ab* can have 5 values. For each of these values *cd* can have 4 values and for each of these 20 values of *abcd*, *ef* have 3 values. Therefore, we can consider a set of 60 numbers.

Consider N = *abcdef* = 100 (*abcd*) + *ef*. As *cd* is a multiple of 8 (and hence of 4), *abcd* is a multiple of 4 and 100 (*abcd*) is a multiple of 16.

$\therefore$ N is a multiple of 16 if and only if *ef* is a multiple of 16 (i.e if *ef* = 16, 32 or 96)

If *ef* = 16, there are 4 values for *ab* and 3 for *cd*, i.e., 12 for *abcd*. Similarly, if *ef* = 32 or 96 there are 12 for *abcd*.

$\therefore$ There are 12 (3) or 36 values of N which are divisible by 16. Ans: (36)

Solutions for questions 6 to 22:

6. Let the initial lengths of the candles be 30 cm and 20 cm respectively. Let their hourly rates of burning be 2*k* cm and *k* cm respectively. So, in three hours (from 6:00 p.m. to 9:00 p.m.) they burn 6*k* cm and 3*k* cm.

Given that $\frac{30-6k}{20-3k} = \frac{2}{3}$

$\Rightarrow 90 - 18k = 40 - 6k$

$\Rightarrow k = \frac{25}{6}$

Now, let them be of equal length, say, *t* hours after 6:00 p.m.

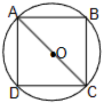
$\Rightarrow 30 - \frac{25t}{3} = 20 - \frac{25t}{6}$

$\Rightarrow 10 = \frac{25t}{6} \Rightarrow t = 2.4$

Any other relationship between A_1, A_2, A_3, A_4 that we can write will not be independent of the above two relationships.
For example, consider the relationship.
$A_4 – A_1 = 3(A_3 – A_2)$
$\Rightarrow (4a_4 - a_1) - (a_1 - 2a_2) = 3[(3a_3 - 4a_4) - (2a_2 - 3a_3)]$
$\Rightarrow 16a_2 – 18a_3 = 2a_1$
$\Rightarrow 8a_2 – 9a_3 = a_1$ ----- (3)
Both the equations (2) and (3) are the same.
Hence, we cannot get a unique value for $a_1 : a_2$.
Choice (D)

$\therefore$ The candles are of equal length at 6:00 p.m. + 2.4 hours, i.e., at 8:24 p.m. Choice (C)

7. The situation is equivalent to inscribing a square in a circle and then inscribing a triangle of the maximum possible area in the square.



Let ABCD be the square inscribed in the circle with centre 'O'. ΔACD is the largest triangle that can be inscribed in the square ABCD. Let AB = BC = CD = AD = a
$\Rightarrow AD^2 + DC^2 = AC^2 \Rightarrow 2a^2 = 16$
$\Rightarrow a = 2\sqrt{2}$

Area of the triangle = $\frac{1}{2} \times a \times a = \frac{1}{2} \times 2\sqrt{2} \times 2\sqrt{2}$

= 4 sq.cm

Choice (B)

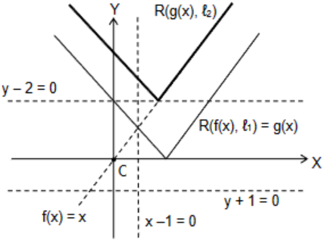
8. Let us suppose everyday Divya crossed the bus at 7 a.m. at point A (say). On the specified day Divya was late, she crossed the bus at 7:05 am at point B (say).
Divya, being 15 minutes late, will reach point A at 7:15. Thus, Divya takes 10 minutes to go from B to A (7:00 to & 7:05).
The bus is on time and so it takes 5 minutes to go from A to B (7:00 to 7:05)
Thus, the speed of the bus is twice that of Divya.
Choice (A)

9. There are 4 cards of each denomination in a pack of 52 cards. Each set of four cards of the same denomination can be distributed among 4 persons in 4! Ways. Hence, the required number of ways is $(4!)^{13} = (24)^{13}$ Choice (B)

10. From the common data, we get the following possibilities for *m* and *n*.

m	*n*
16	19
81	41

12. R ($f(x)$, ℓ_1) is shown in ordinary type. R($g(x)$, ℓ_2) is shown in bold.



We see that it has a minimum value of 2.
Choice (C)

13. From statement (I) alone: 1st of this month is a Sunday and 1st of next month is Wednesday i.e., the gap between 1st of this month and 1st of next month is 3 odd days. So, this month has 31 days, i.e., this month can be January, March, May and so on. Hence, Statement I alone is not sufficient.

From statement (II) alone: 1st of previous month is a Saturday and 1st of this month is a Sunday i.e., number of odd days is one. So, the previous month has 29 days i.e., it is February, and this month is March. Hence we can find the day on which 1st of January falls.
Hence, II alone is sufficient. Choice (A)

14. $(12)_n = n + 2$
$(34)_n = 3n + 4$

I. $m < n$, if the ten's digit of m is less than its units digit, i.e. if $m = 16$, $m < n$. True.

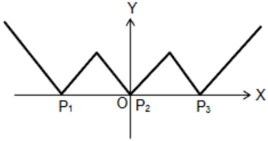
II. $| m - n |$ is a perfect square could be true, as $|16 - 41|$ is a perfect square.

III. $m + n$ is a perfect square could be true, as $81 + 19 = 100$.

None of the statements is definitely false.

Choice (D)

11. In the first operation we get two points at which R(*f*(*x*),1) touches the x–axis. For every subsequent operation one more point gets added. Hence after two operations there will be three points at which R(R(*f*(*x*), ℓ), ℓ) touches the x–axis. The graph will look as follows, where P_1 through P_3 are the three points.



Ans: (3)

- - - - -

$[(2m - 1)(2m)]_n = (2m - 1)n + 2m$

$\therefore X = (1 + 3 + 5 + \ldots(2m - 1))n + (2 + 4 + 6 + \ldots 2m)$

$\therefore X = nm^2 + m(m + 1)$

$= (2m + 1)m^2 + m^2 + m = 2m^3 + 2m^2 + m$

$\therefore X + m + 2 = 2m^3 + 2m^2 + 2m + 2 = (2222)_m$

Alternative solution:

Since it is given that $m > 2$, without loss of generality, say $m = 3. \Rightarrow n = 7$

$\therefore X = (12)_7 + (34)_7 + (56)_7 = 7(1 + 3 + 5) + (2 + 4 + 6) = 75$

$\therefore X + m + 2 = 75 + 5 = 80$

Now, $(80)_{10} = (3^3 \times 2) + (3^2 \times 2) + (3 \times 2) + (2) = (2222)_3$.

Ans: (2222)

15. $\log x + \log 6 = \log(x \times 6) = \log 6x$.

$\log(x + 6) = \log 6x \Rightarrow x + 6 = 6x \Rightarrow x = 6/5 = 1.2$

Choice (B)

16. $(2x^2 + 13)(x - 3) + (2x^2 + 13)(x - 5) = 0$

$\Rightarrow (2x^2 + 13)(2x - 8) = 0$

$\Rightarrow 2(2x^2 + 13)(x - 4) = 0$

$\Rightarrow (2x^3 - 8x^2 + 13x - 52) = 0$

Therefore, the product of the roots of the above equation

$= -\left(\frac{-52}{2}\right) = 26$

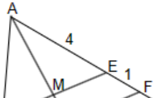
Alternative Solution:

The product of the roots of any polynomial of degree $n = (-1)^n \left(\frac{\text{constant term}}{\text{coefficient of } x^n}\right)$

Given $n = 3$, we only need the leading coefficient $(2 + 2 = 4)$ and the constant $((13) \times (-3) + (13) \times (-5) = -104)$. Hence required product $= -\left(\frac{-104}{4}\right) = 26$.

Choice (A)

17.



Then,

$p + rq = 25$ - - (1)

$q + rp = 17$ - - (2)

(1) – (2) gives,

$(p - q) + r(q - p) = 8$

$\Rightarrow q - p = \frac{8}{r - 1}$

Now, $q - p$ is a positive integer i.e., $\frac{8}{r - 1}$ is a positive integer

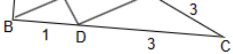
$\Rightarrow r = 2, 3, 5$ or 9 - - (3)

(1) + (2) gives,

$(p + q) + r(p + q) = 42 \Rightarrow p + q = \frac{42}{r + 1}$

Now $p + q$ is an integer

So, $\frac{42}{r + 1} \Rightarrow r = 1, 2, 5, 6, 13, 20$ and 41 must be an



Through D, we draw a line parallel to BE to intersect AC at F. Now, in ΔBEC, BD : DC = 1 : 3 (given) and BE ∥ DF.

∴ EF : FC = 1 : 3

And AE : EC = 1 : 1 = 4 : 4

∴ AE : EF = 4 : 1 and in ΔADF, ME ∥ DF. Hence, AM : MD = 4 : 1 Choice (D)

18. Let the time taken by Ashok to complete the job be x hours. The standard time would be $\frac{3x}{4}$ hours.

Let y be the standard rate of pay.

∴ Total work = standard time × standard rate of work

$= \frac{3x}{4}y = \left(\frac{3}{4}y\right)x$

∴ Ashok's rate of work $= \frac{3}{4}y$

∴ Ashok's actual rate of work was 25% less than the standard rate.

∴ His actual hourly wage $= 6 - \left[\frac{\frac{25}{5}\times 8}{100}\times 6\right] = 3.6$

Since he took exactly three hours, he was paid 3.6 × 3 = ₹10.80 Choice (C)

19. Here, $a = 2{,}000$, $d = 500$

Checking for $\frac{n}{2}(2a+(n-1)d) \le 2{,}00{,}000$

$\frac{n}{2}(4000+(n-1)500) \le 2{,}00{,}000$

$\Rightarrow 40n + 5n^2 - 5n \le 4000$

$\Rightarrow n^2 + 7n - 800 \le 0$

$\Rightarrow n = 25$ i.e. amount paid in 25 instalments

∴ The 25th instalment = 2000 + 24 x 500 = 14,000

Ans: (14000)

20. Let the number of direct roads from A to B be p, B to C be q and C to A be r.

integer.

∴ from (3) and (4), r = 2 or 5

If r = 2, (1) ⇒ p + 2q = 25 and (2) ⇒ 2p + q = 17

∴ p = 3 and q = 11

If r = 5, (1) ⇒ p + 5q = 25 and (2) ⇒ 5p + 1 = 17

∴ p = 2.5 and q = 4.5. As p, q are integers, r = 2, p = 3 and q = 11. Ans: (2)

21. Let the time taken to complete the work when all the three work together be *t* days. Hence Amar, Akbar and Anthony take *t* + 20, *t* + 8 and 2*t* days respectively to complete the work.

Hence, $\frac{1}{t} = \frac{1}{t+20} + \frac{1}{t+8} + \frac{1}{2t}$

Considering the given options, the above equation is satisfied when *t* is 4. Choice (B)

22. $f(x) = \frac{25^x}{25^x+5}$

$\therefore f(1-x) = \frac{25^{1-x}}{25^{1-x}+5} = \frac{5}{5+25^x}$

Thus, $f(x) + f(1-x) = \frac{25^x}{25^x+5} + \frac{5}{25^x+5} = 1$

$\therefore f\left(\frac{1}{99}\right) + f\left(\frac{98}{99}\right) = 1$

$f\left(\frac{2}{99}\right) + f\left(\frac{97}{99}\right) = 1$

$f\left(\frac{49}{99}\right) + f\left(\frac{50}{99}\right) = 1$

Therefore adding,

we get $f\left(\frac{1}{99}\right) + f\left(\frac{2}{99}\right) + \ldots f\left(\frac{98}{99}\right)$

= 49(1) = 49 Ans: (49)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	15
Easy	7, 10, 13, 21
Medium	1, 3, 4, 6, 8, 14, 16, 17, 18, 19
Difficult	2, 5, 9, 11, 12, 22
Very Difficult	20

Done

Tip: Scroll slowly for the best results